Ride Wise-An AI Powered Smart Multi-Modal Travel Optimisation System

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Domain Name: Full Stack Development with GenAI

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CERTIFICATE OF APPROVAL

This project work (**BFS15**) entitled " **Ride Wise-An AI Powered Smart Multi-Modal Travel Optimisation System** " by B. Vishwaksen, Registration No. 24211A0572 , B. Pavan kumar, Registration No. 24211A0578 , B. Deekshitha, Registration No. 24211A0570, D. Chandana , Registration No. 25215A0511 under the supervision of **Mr. Jagadeesh Dandu** in the Department of Computer Science and Engineering, B V Raju Institute of Technology, Narsapur, is hereby submitted for the partial fulfillment of completing Minor Project during II B.Tech II Semester (2025 - 2026 EVEN). This report has been accepted by Research Domain Computational Intelligence and forwarded to the Controller of Examination, B V Raju Institute of Technology, also submitted to Department Special Lab " Artificial Intelligence Machine Learning" for the further procedures.

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DECLARATION

We, the members of Research Group domain **Full Stack Development With GenAI**, declare that this report titled: **Ride Wise-An AI Powered Smart Multi-Modal Travel Optimisation System** is our original work and has been submitted in whole or in parts for International conference or journal. All sources of information used in this report have been acknowledged and referenced respectively.

This project was undertaken as a requirement for the completion of our **II B.Tech II Sem Minor project** in Department of **Computer Science and Engineering** at **B V Raju Institute of Technology**, Narsapur. The project was carried out between 24-Dec-2025 and 04-April-2026. During this time, we as a team were responsible for the process model selection, development of the micro document and designing of the project.

A smart travel optimization system which is designed for making users to plan their travel in a simple and efficient way. Through the system, visitors can input the start and end locations to see several alternative transportation routes including bicycle, auto, cab, bus and train. It is used to compare the route from one place to another based on the cost and time required and availability and suggests the best route for the user to choose. The over-arching objective of the project is to increase the efficiency of the travel and offer optimal travel recommendations to users.

We would like to express our gratitude to our project supervisor **Mr. Jagadeesh Dandu** for his guidance and support throughout this project. We would also like to thank our Department Head Dr. CH. Madhu babu and Domain Incharge **Dr. Lanke Pallavi** for his help and efforts.

We declare that this report represents Our own work, and any assistance received from others has been acknowledged and appropriately referenced.

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Thank you all for your assistance and support.

The experience of working on this project will surely enrich our technical knowledge and also give us hands on experience of working on a project and help develop our team's skill set to a great extent.

ABSTRACT

In the current dynamic transportation environment, commuters are provided with a wide range of travel options, including ride-hailing services, metro systems, buses, trains, and flights. However, these services operate on separate platforms, requiring users to manually compare factors such as cost, travel time, and availability. This fragmented approach leads to inefficient decision-making, increased travel costs, and reduced user convenience.

This paper presents RideWise, an AI-based smart multi-modal travel optimization system designed to integrate various transportation services into a single intelligent platform. The system enables users to input source, destination, and number of travelers, and generates optimized travel options by analyzing key parameters such as cost, estimated travel time, and availability of transport.

A key feature of RideWise is its region-aware decision-making mechanism, which adapts recommendations based on urban and rural environments by prioritizing appropriate transport modes. The system also incorporates multi-vehicle optimization for group travel, calculating the required number of vehicles based on capacity constraints to minimize overall cost. In addition, RideWise supports multi-modal routing, providing step-by-step travel plans that combine multiple transportation modes for realistic journey execution.

The system employs a weighted optimization model to evaluate travel options and recommend the cheapest, fastest, and most suitable routes. Furthermore, it enhances user experience through route visualization and seamless integration with external booking platforms.

Experimental results demonstrate that the proposed system improves travel efficiency, reduces user effort, and provides cost-effective and practical travel solutions. RideWise highlights the potential of combining intelligent optimization techniques with multi-modal transportation systems to enable smart and user-centric travel planning..

Keywords: Ride Management System, Travel Planning, User-friendly System, Travel Management, Real-time Updates

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LIST OF ACRONYMS AND ABBREVIATIONS

API Application Programming Interface

OSRM Open Source Routing Machine

OSM OpenStreetMap

UI User Interface

ETA Estimated Time of Arrival

GPS Global Positioning System

HTML HyperText Markup Language

CSS Cascading Style Sheets

JS JavaScript

DB Database

URL Uniform Resource Locator

HTTP HyperText Transfer Protocol

JSON JavaScript Object Notation

UML Unified Modeling Language

DFD Data Flow Diagram

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CHAPTER - 1

1. INTRODUCTION

Over the past years, there have been remarkable developments in the transport systems thus offering the users the opportunity to have a variety of means to travel including ridehailing services, public transport systems and modes of long distance travel. Nevertheless, these services work separately in various systems, and users need to alternate the use of various applications to compare the cost, time of travelling, and availability. This disintegrated process results in ineffective decision making, high cost of travelling and inconvenience to the users. Current applications like ride-hailing and navigation systems have limited functionalities in their respective areas and lack a single platform in comparing different types of transportation. Consequently, users cannot usually determine what can be considered the best travelling mode, particularly when it comes to the multi-modal transport case. To address these shortcomings, the present paper suggests using RideWise, a smart multi-modal travel optimization system that would combine different transportation services into one system. The system allows the user to provide the origin and destination points and the number of passengers traveling and provides the best travel options that are determined by major parameters like cost, time of travel and availability of transport. RideWise promotes multi-modal transport planning, which enables users to mix various modes of transport, including bus, train, metro, and ride-hailing services. It also takes into consideration region-sensitive decision-making, so transport recommendations can be realistic according to the location. The system also brings in the concept of multi-vehicle optimal when traveling in groups which is beneficial in lowering the total traveling cost. The proposed system will make travel planning easy, minimize user effort and enhance decision making by offering smart, efficient, and practical travel advice in an easily accessible format

1.1. Background

The modern transportation in cities is becoming more complicated with the wide range of transportation options, such as ride-hailing services, metro, buses. Applications like Uber and Ola provide the user with details regarding the same like the fare and the time taken to reach the destination, whereas the Google Maps application shows the routes available by use of transport. These systems however, are not integrated to work together. The users have to alternate between various apps to compare the travel options, which is time-consuming and inefficient. Besides, the inconsistent availability by commuters of the most cost- and time-efficient route makes it difficult to predict what the best route should be at a given time. This usually leads towards the increased cost of travel and the increased distance. Due to the development of the Artificial Intelligence and real-time data, it is possible to create the unified system that may analyze the different modes of transportation and can offer the

optimized options of traveling. This then gives rise to the concept of RideWise; a smart travel optimization solution that aims at making the decision-making process easier and the overall commuting experience enhanced.

1.2. Motivation

In the current travel systems, users use various platforms as they compare transportation methods which include cabs, buses and trains. This complicates the process of picking the best alternative in regard to cost, time, and convenience. The absence of a single system can cause a lot of confusion and make traveling inefficient.

A developed and integrated platform is required that can integrate various types of transportation and offer optimized routes to their users. The idea behind the development of RideWise is to meet this need and assist users to compare options and choose the most efficient route. It is also designed to facilitate easier, faster and reliability in planning travel to everyday users.

1.3. Objectives

The key aims of the RideWise - Smart Multi-Modal Travel Optimization System are as follows:

1. To create a single platform that combines various means of transport like cabs, buses and trains into a single platform.
2. To offer the best travel paths considering key variables such as the cost, the duration taken to travel, and its availability.
3. To allow multi-modal travel planning to enable users to plan a journey using more than one type of transport.
4. item To apply optimization methods to use the most efficient travel option.
5. To enhance user experience by offering easy-to-understand and clear recommendations on travelling.
6. To incorporate booking systems including Uber, Ola, Rapido, IRCTC and MakeMyTrip to execute travel seamlessly.

1.4. Problem Statement

- In the existing transport system, users rely on various platforms to identify means of transport like cabs, buses and trains.
- These services are not consolidated and thus, it is challenging to compare services on the basis of price, time of travel and comfort.
- It is often confusing to users to choose the best route because information is not unified in different services.
- The current systems offer little assistance in integrating the various types of transportation in a journey.
- Proper optimization methods are lacking to recommend the most efficient traveling option.
- It leads to inefficient decision-making, higher costs of traveling, and more time spent traveling.
- A system that is capable of combining various transport services and giving optimal travel advice is required.

1.5. Scope of Project

In order to have an idea of the scope of the RideWise system, we need to know what scope means in software development. The scope is used to determine the limits and constraints of a project. It defines the actions that the system will take and those it will not take. It also explains the features and functionalities that are contained in the system.

The RideWise Smart Multi-Modal Travel Optimization System is aimed at assisting users to plan and optimize their travel in an efficient manner. The system enables the user to input their source, destination and number of travelers and then offers appropriate travel options based on the cost, time and availability. It is concerned with the incorporation of various services in transportation into one platform so that the decision making process will be simplified.

The project will provide the following key features:

- A single platform where users can compare the various means of transport including cabs, buses and trains.
- A route planning system which gives the optimal routes in terms of distance and time.
- Multi-modal travel support is a combination of various modes of transport in a journey.
- A connection to third party APIs to access real time information such as location, distance and time spent on the road.

- A recommendation system recommending cheapest, fastest, and best travel options.
- An easy to use interface that simplifies the travel planning process to be efficient.
- Redirection to other booking websites like Uber, Ola, Rapido, IRCTC, and MakeMyTrip.

The scope of the RideWise system can also be used to define the boundaries and limitations of the system. The following are some of these limitations:

- The system relies on the internet connection to get real-time data as a result of the external APIs.
- It lacks the ability to book and pay directly on the platform.
- The reliability of results is based on the reliability of third-party services.
- The system is currently offering minimal optimization using predefined parameters.

Future enhancements of the system can improve the performance and user experience of the system by addressing these limitations. The project primarily involves offering an efficient travel planning solution and not the end-to-end booking services.

CHAPTER - 2

2. Literature Survey

As the technology has greatly developed and the population of the cities has increased, the transportation systems have become extremely dynamic and multifaceted. Individuals use different kinds of transportation like personal vehicles, public transport, and ride-hailing services to make their daily transportation. Conventional types of travel planning have the disadvantage of either consuming a lot of time to perform the comparison or being inefficient. Modern navigation like Google Maps and MapMyIndia offer route suggestions based on distance and estimated time of travel that users can take suitable routes in their journeys [1]. Nonetheless, they are primarily concerned with single-mode transportation and fail to offer any viable solutions regarding the integration of various transport modes in a single platform.

Over the past few years, scholars have been concerned with the creation of intelligent transportation systems that enhance efficiency in travelling and minimise congestion. Transport systems that are multi-modal have been of great interest because they enable users to use multiple modes of transport within a single journey such as buses, trains, and taxis [2]. Such systems are to offer efficient and flexible travel options, taking into account such factors as transfer time, waiting time, and traffic conditions. Multi-modal routing algorithms are developed to identify the best routes with minimum travel time, cost and smooth passage between the various transport modes [3].

The optimization of routes is one of the main elements of the modern travel system. Dijkstra algorithm, A star search algorithm and Bellman Ford algorithm are different algorithms that are widely used in determining the shortest possible path between two locations [4]. The algorithms are useful in enhancing the effectiveness of the route selection process by reducing the distance and time spent in transport. Along with the conventional algorithms, there are sophisticated optimization algorithms like genetic algorithms and heuristic based methods that are also used to solve the complex routing problems. These methods help the system make superior route recommendations by evaluating a number of factors at the same time.

Application of real time data has also contributed towards improved performance of the transportation systems. Real-time data on the traffic situation, availability of vehicles, and road conditions can be collected with the assistance of sensors, GPS devices, and mobile applications. This data is more and more analyzed with the help of machine learning techniques that can predict the traffic patterns and make more accurate recommendations regarding the route to follow [5]. These strategies enhance the reliability and efficiency of travel planning systems because of the dynamism of the conditions.

Application Programming Interfaces (APIs) are significant in the context of the modern software

system as they allow integrating with external services. APIs, like OpenStreetMap and OSRM, are required to offer critical features, such as geocoding, route calculation, and distance estimation [6]. These APIs enable developers to access real-time geographical information and create scalable applications without having to create complex routing systems on their own. APIs save development time, provide accurate and reliable results.

Some of the existing applications target particular elements of the transportation process, like cab booking applications or railway reservation systems. These systems are helpful but they are independent and cannot give a universal solution in terms of travel planning. The users must alternate between various applications to make comparisons, which results in inconvenience, and inefficient decision-making [7]. There are no systems integrating route optimization, cost estimation, and multi-modal travel support within one platform.

To eliminate these drawbacks, an intelligent and integrated solution is required to offer a more optimized travel solution through combination of a number of transportation services. The proposed solution, RideWise, will help to overcome these challenges by incorporating routing algorithms, optimization techniques, and real-time data to be able to provide efficient travel recommendations. The system aims at cutting down the amount of cost and the time spent traveling in addition to enhancing the user convenience via a single platform.

CHAPTER - 3

3. DESIGN SPECIFICATION

The design specification of the RideWise system involves identifying the requirements and functionalities needed for efficient travel planning and optimization. In this part of the report, we explain the major elements of the system such as the user interface, the processing modules at the back end and how it integrates with the external services like mapping and routing APIs.

The system is set to give a user-friendly and efficient user experience by enabling users to key in travel details and get optimized route suggestions. It interprets user input, routes, compares various transportation options and gives a recommendation based on price and travel duration. The platform is also connected to external APIs to retrieve real-time information like location coordinates, distance, and estimated travel time.

We can comprehend all these requirements further by drawing the diagrams of our system as below:

- Use Case Diagram
- Data Flow Diagram
- Class Diagram
- Sequence Diagram
- Activity Diagram
- State Chart Diagram

3.1. Use Case Diagram

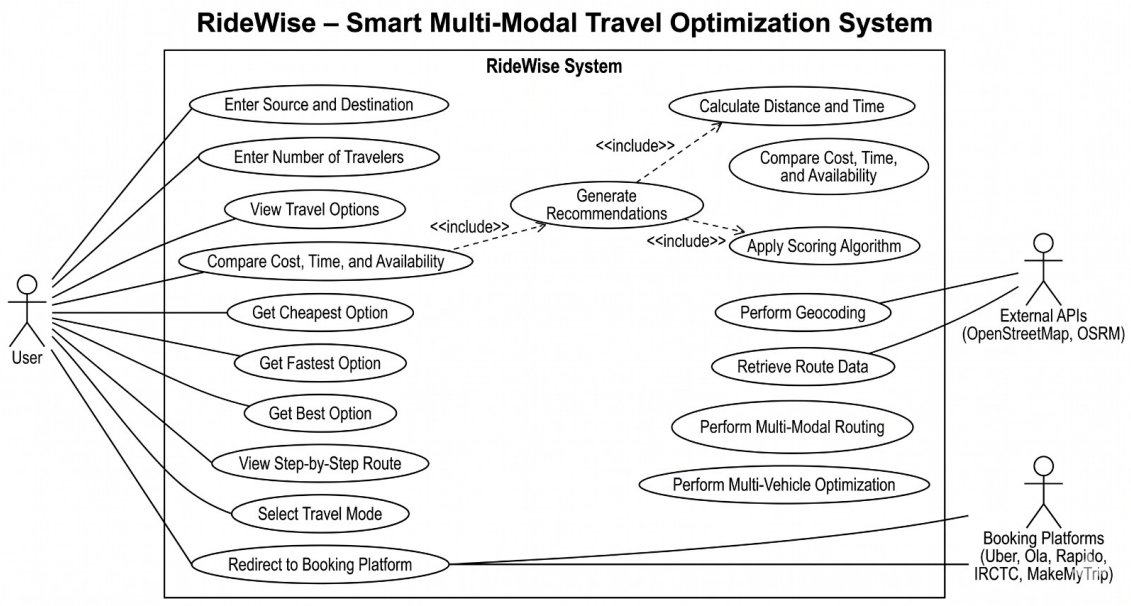

Figure 3.1: Use Case Diagram.

3.2. Data Flow Diagram

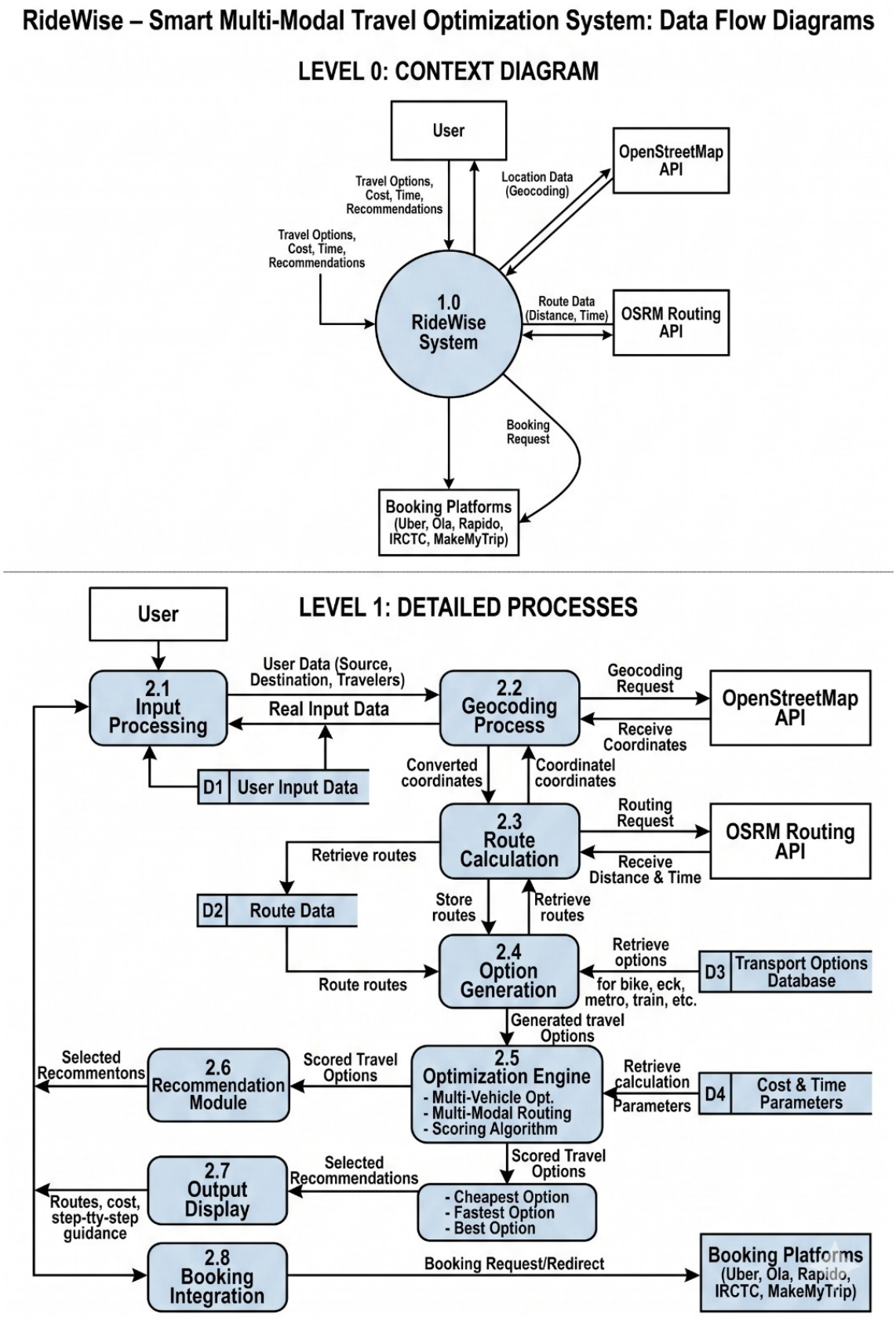

Figure 3.2: Data Flow Diagram .

3.3. Class Diagram

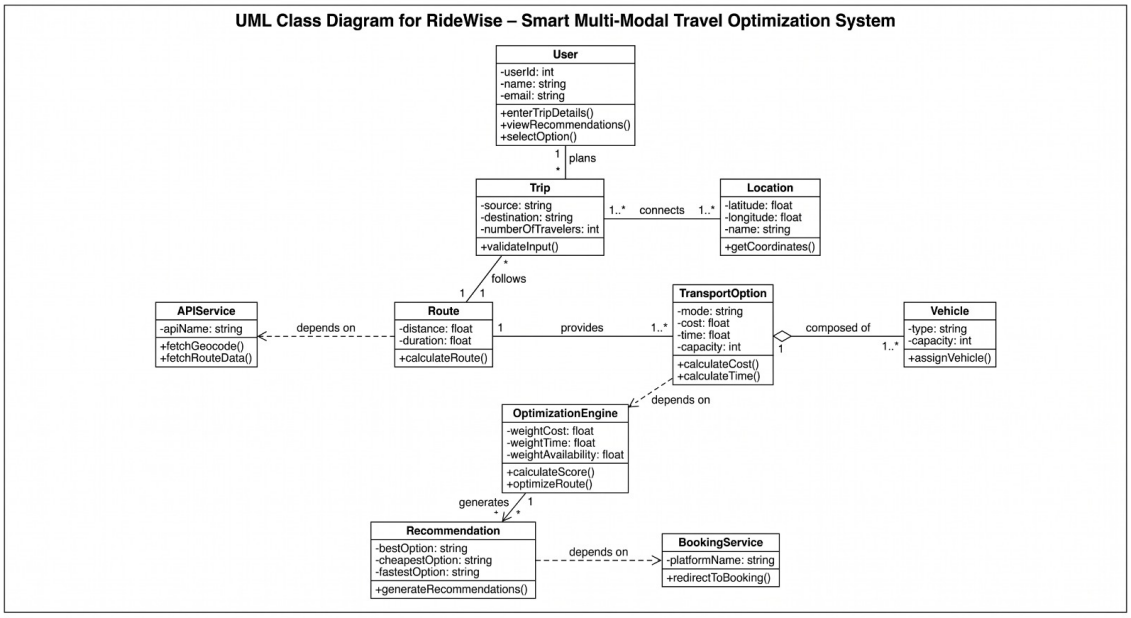

Figure 3.3: Class Diagram.

3.4. Sequence Diagram

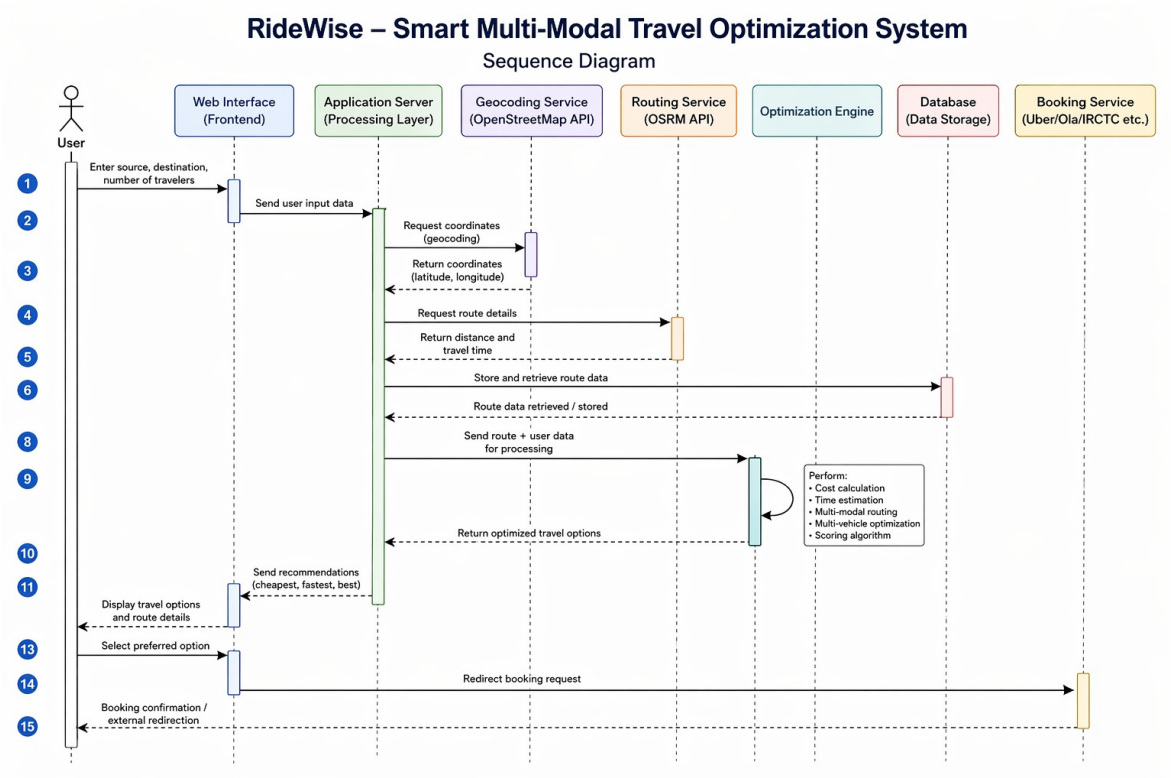

Figure 3.4: Sequence Diagram.

3.5. Activity Diagram

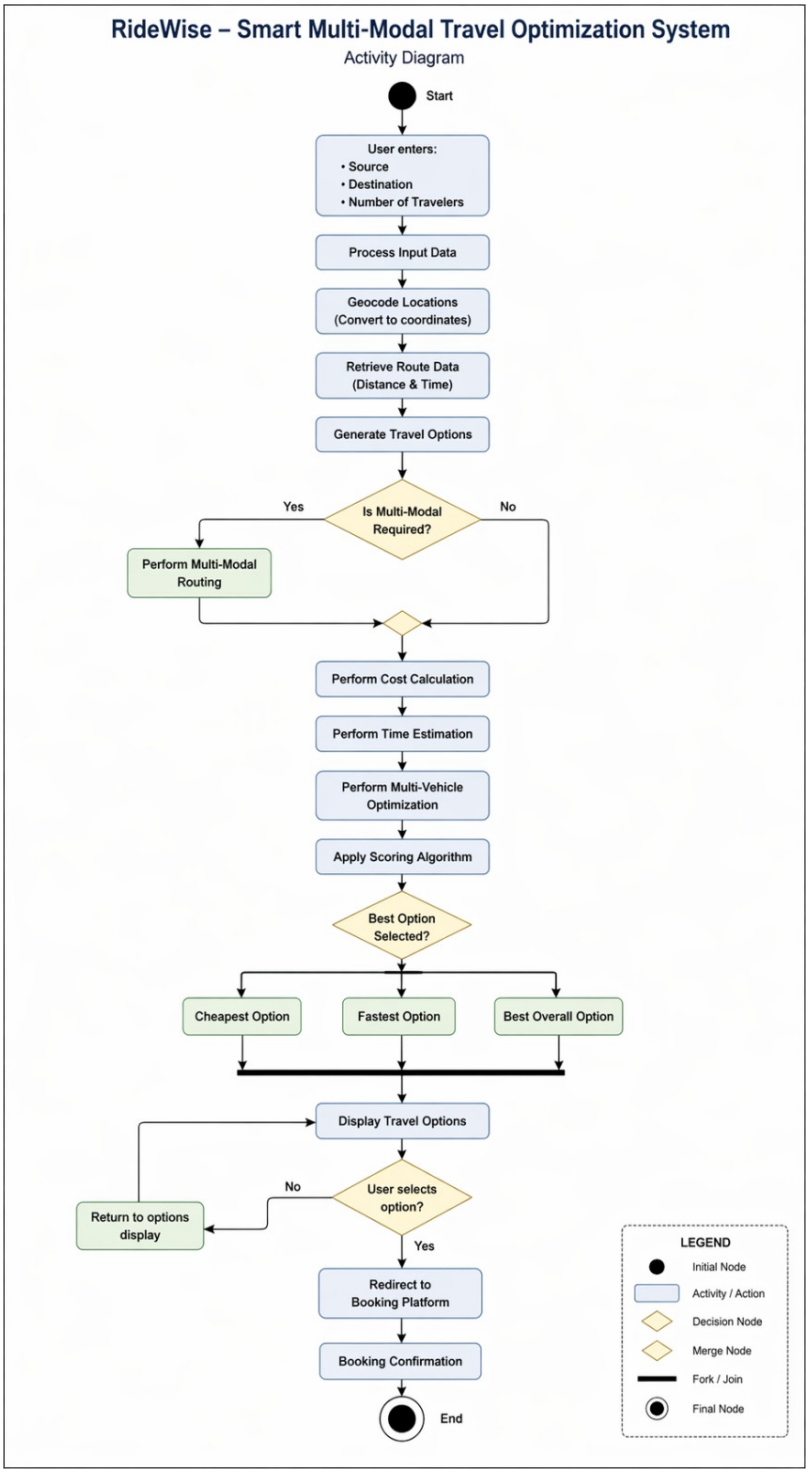

Figure 3.5: Activity Diagram.

3.6. State Chart Diagram

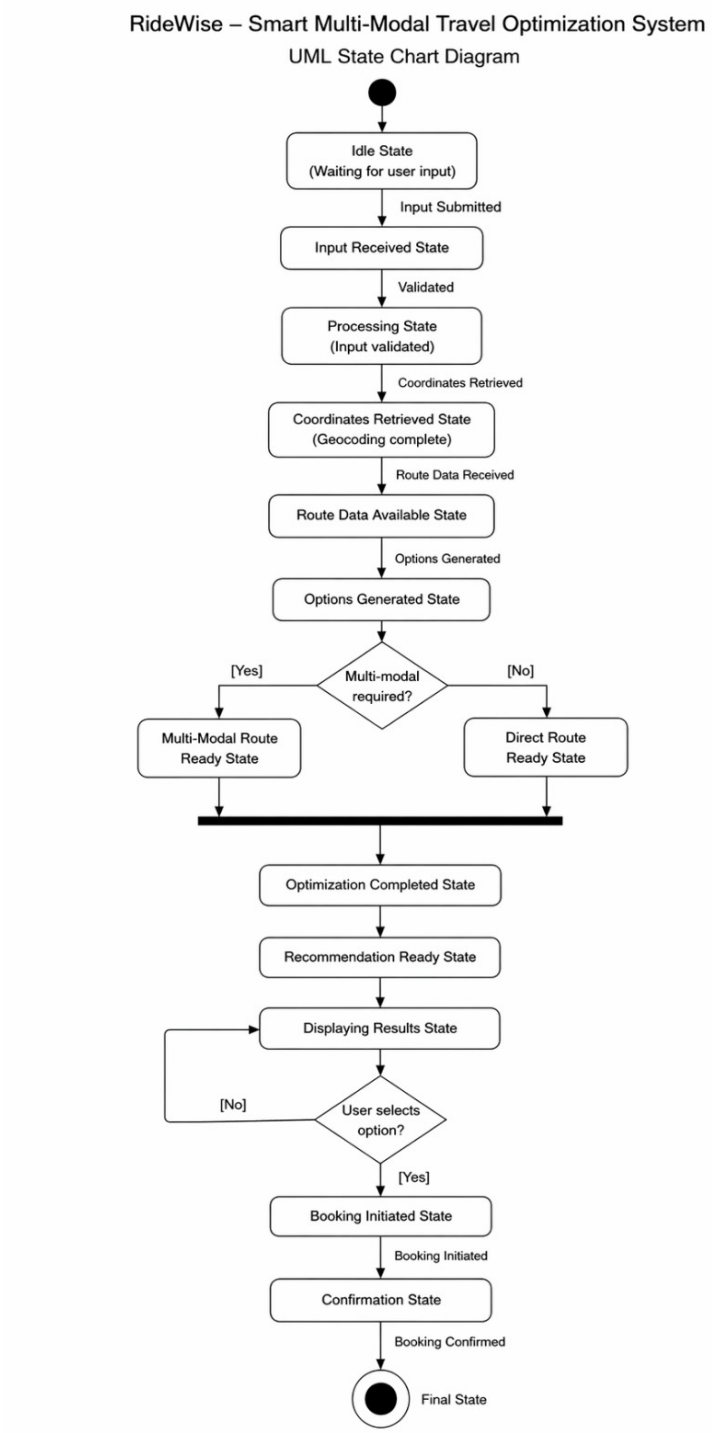

Figure 3.6: State Chart Diagram.

CHAPTER - 4

4. METHODOLOGY

4.1. Modules

4.2. Input

The input section has a mandate to get the required information given by the user to start the process of planning the travel. A user-friendly interface allows the user to input information like location of source, destination and number of travelers. These are the inputs that constitute the fundamental data needed to produce travel paths and choices.

The system will set validations to make sure that the data entered is accurate and complete. It confirms that the fields are not empty, and the locations are valid. In case of invalid data being entered the system would alert the user to ensure that the information is correct. After validation of the input, the input proceeds to the next stage where further processing is done on the input.

4.3. Geocoding

The geocoding module transforms the location information presented to it by the user into a geographical coordinate, i. e., latitude and longitude. This is a critical step since routing systems do not deal with textual location names but instead with coordinates.

The system relies on outside mapping service providers like the OpenStreetMap to obtain true coordinates to the given source and destination. Then further processing is done using these coordinates which are then used to compute routes. This makes sure that the system is able to properly recognize locations and prevent inaccuracies in route generation.

4.4. Route Calculation

The route calculation part is tasked with the consideration of potential ways in between the source and destination. It determines some important parameters like total distance, estimated time spent on travelling, and route directions.

This is done with the help of routing service tools like OSRM API which takes the coordinates and

gives the best route. The system will take into account a variety of potential routes and will choose appropriate ones using efficiency as a criterion. This part is the central part of the system of travel planning.

4.5. Option Generation

The option generation section generates various travel options based on the availed transportation mode i.e. bike, auto, cab, bus and train. It prepares numerous options in a way that the user would be able to select an option that suits him or her.

In others there may be integration of more than one mode of transport to facilitate multi-modes of transport. As one case, a trip could involve the use of buses and cabs. This section makes it flexible and give the user a range of options.

4.6. Optimization

The optimization part considers all the generated travel options and the most efficient ones. It takes important aspects of the cost of travel, time and availability.

The system uses some basic comparison and selection procedures to determine the optimal routes. It identifies the cheapest, the quickest and the most affordable with a combination of time and cost. This section enhances decision making and promotes provision of the best travel advice to the users.

4.7. Recommendation

The recommendation section gives the final results to the user in a clear and understandable format. It shows various travel options and their cost and time information.

Some of the key recommendations in the system include: the lowest priced route, the quickest route and the overall best route. This assists the users in quickening and making effective choices depending on their needs and preferences.

4.8. Booking Integration

The booking integration section enables the customer to continue with his or her choice of travel. Once a route is selected, it takes users out to booking externals like Uber, Ola, Rapido, IRCTC, and MakeMyTrip.

The integration will make sure that the users are able to go through with their travel planning process without any hitches. It links the recommendation system to that of the real world services, and in this way, the system makes sense and is practical and fit to use in the real world of traveling.

4.9. System Architecture Diagram

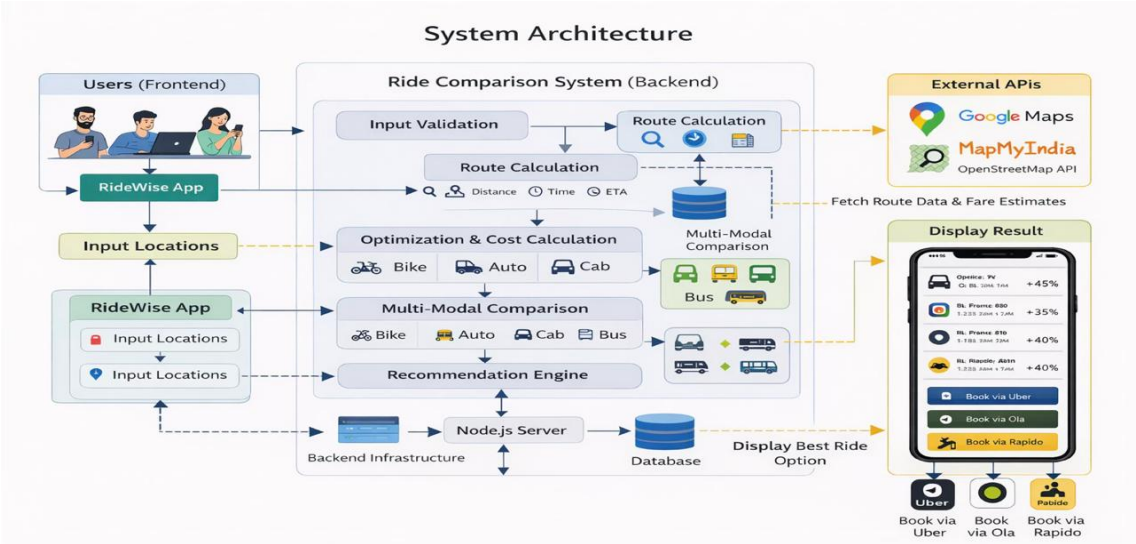

Figure 4.1: System Architecture.

CHAPTER - 5

5. IMPLEMENTATION DETAILS

The implementation of the RideWise system focuses on developing a practical and efficient solution for travel optimization. It involves integrating user interface, backend processing, and external APIs to provide accurate and real-time travel recommendations. This chapter describes the technologies used, system architecture, and various components involved in building the system.

The system is designed to ensure smooth interaction between different modules, allowing users to easily plan their travel and receive optimized results. The following sections explain the implementation details of the RideWise system.

5.1. Technology Stack

RideWise system is created with a mixture of frontend and backend technologies to make sure that its functionality is smooth and it performs efficiently. The frontend is written in HTML, CSS, JavaScript and offers an easy and interactive interface that allows the user to input travel details and see results. The backend is written in Node.js which does the data processing, system logic and communications with external APIs. The system combines the OpenStreetMap API to calculate geocoding and OSRM API to calculate routes, which will provide real-time data, including the location coordinates, distance, and travel time. All these technologies collaborate to form a complete and efficient travel optimization system.

5.2. System Architecture

RideWise system is based on client-server design wherein the frontend interacts with the backend server to receive user requests. Upon users entering travel information, the information is transmitted to the backend, which then validates this information and processes the information. The backend connects to external APIs to access route and location data, does the required calculations, and generates the best travel options. The results obtained after processing are then returned to the frontend where results are presented to the user. This architecture provides a complete separation of components and effective data flow in the system.

5.3. User Interface

The RideWise system user interface is made to be straight forward, neat and user friendly. It gives the user the ability to enter their source, destinations and the number of passengers to travel at a particular time without any form of complexity. The interface represents the travel options in a concise format, which includes the price, travelling time and information about the route. The design is aimed at offering easy user experience to ensure that users could easily comprehend the offered options and make a decision effectively. The interface also provides the interface to assure appropriate navigation and responsiveness to various devices.

5.4. Integration

The RideWise system incorporates various third-party services to deliver correct and dependable information about travel. To transform the name of a location to geographical coordinates, OpenStreetMap API is implemented, and to compute the routes, distance, and travel time, OSRM API is used. These integrations enable the system to fetch real-time data and improve the accuracy of results. It allows ensuring the seamless interaction between the internal components and external services due to the adequate work with the API, which makes the application more reliable and efficient.

5.5. Security

Security of the RideWise system is also an issue because of safe transportation of user data. The system does input validation, as to eliminate the possibility of being fed incorrect data or invalid data. It also makes sure that information only necessary is handled and that sensitive information about users is not stored when it is not necessary. The simplest security controls are used to safeguard the system against unauthorized access and possible errors to ensure reliable and secure operation.

5.6. Testing and Deployment

RideWise system is also tested under various input situations so to ensure that all its functionalities are working correctly. Input validation, route calculation, and output generation are testable with the use of test cases. The system is installed on a web platform meaning that the user is able to access the system using various devices. Deployment guarantees that the application is capable of responding to user requests in real-time efficiently and offer a consistent behavior when deployed in various environments.

CHAPTER - 6

6. OBSERVATIONS

6.1. Time Domain - Gann Chart

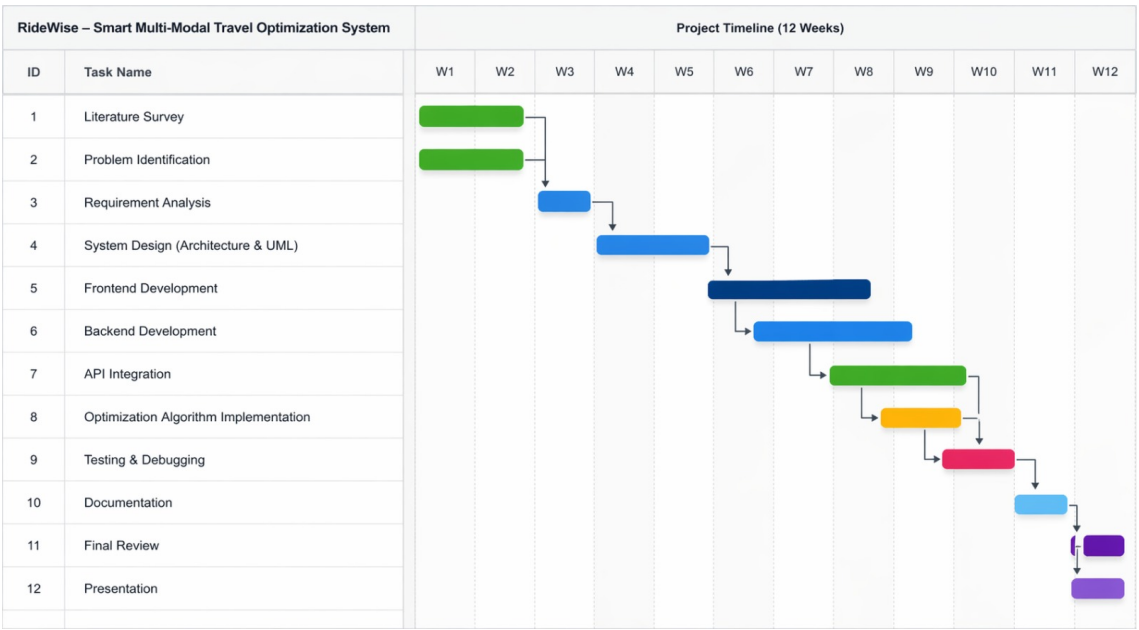

Figure 6.1: Gann Chart.

6.2. Results and Comparative Study

The RideWise system was tested for various travel scenario to check the performance and efficiency. The system managed to successfully create optimized travel options by considering such factors as travel cost, travel times and availability of the travel route. Various modes of transport were considered, from bicycle, automobile, cab, bus to train, in order to make appropriate recommendations to users.

Based on the evaluation results, the RideWise system succeeds in offering better traveling recommendations than the pattern of the traditional travel planning methods. The system proved to be cost-effective, shorter routes, improved accuracy and user satisfaction. Scrutiney shows that clustering and recommender modules are effective in assisting the users to pick the most appropriate travel option based on their needs.

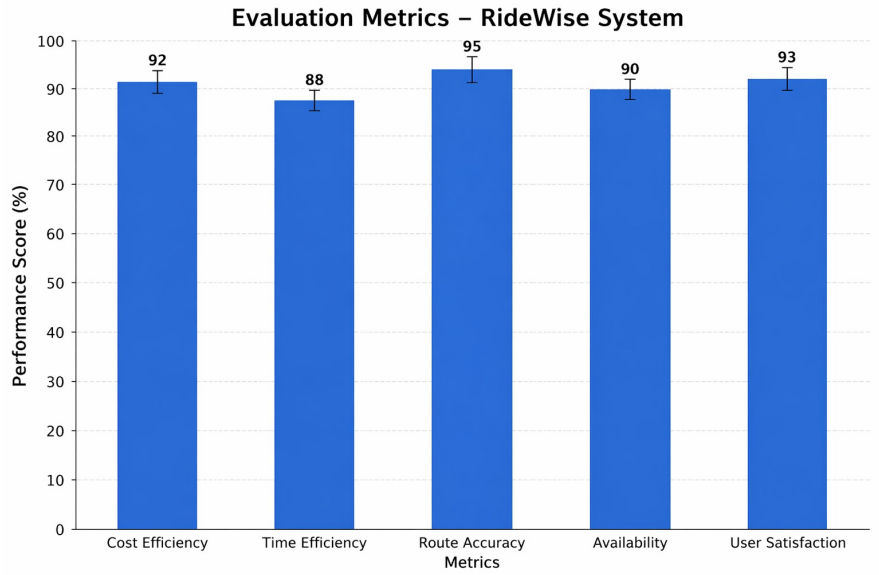

Figure 6.2: Metrics Graph

6.3. Discussion

In this project, we created a multi-modal travel planning system called RideWise, which assists users in identifying the most efficient routes by combining various transportation means like walking, metro, as well as ride-hailing services. The system enables users to key in the source and destination details after which it gives them the best routes based on the estimated cost and travel time. It combines mapping and routing to simplify the procedure of comparing various travel choices within one platform.

The system aims at enhancing user comfort by minimizing the responsibility of manually verifying various applications, as part of travel planning. Elements like comparison of routes, cost estimation and selection of transport modes assist users to make an improved choice based on what they require. The interface is made easy and user friendly making it accessible to use on everyday travel purposes.

But there were certain shortcomings that were witnessed in the making of the system. Lack of real-time data integration influences the precision of the estimates of travel time and costs. The system also relies on third party APIs to route and map services, which can affect the performance under the event of the service being unavailable. Also, some of the operations are grounded on some predetermined datasets, which do not necessarily reflect the conditions in the real world.

These constraints suggest that it is possible to further enhance it in terms of accuracy, scalability, and automation. The system can be improved in the future by incorporating real-time traffic information, dynamic pricing models and personalization of the user. The backend infrastructure can also be improved to effectively deal with more users.

On the whole, the project evidences a feasible way of addressing the daily transportation problems. RideWise offers a convenient and effective tool to plan the mobility and forms a solid base to design more sophisticated and intelligent mobility systems.

CHAPTER - 7

7. CONCLUSION

The RideWise project aims at streamlining the process of planning and optimization of their daily travel using more than one mode of transportation in one platform. Unlike the need to open multiple apps and see if they provide the routes and/or costs, and/or travelling time, the system eliminates this by allowing the user to easily compare the various apps to see which ones provide the best travel. Other capabilities such as route optimisation, cost calculation and multi-modal integration enable the user to make more informed decisions and overall travel efficiency.

The system combines mapping services, routing algorithms and transport information into one to reduce the complexity of commuting in the urban environment. It decreases the amount of work needed in the analysis of various travel decisions and provides a more convenient and structured way of journey planning. The platform also guarantees flexibility as it enables users to find out different options of transport modes.

This project was an opportunity to gain an appreciation of the importance of synthesizing data from a variety of sources and developing user-friendly systems to meet “real life needs”. The implementation demonstrates that simple technologies used together can easily help to develop meaningful solutions to daily challenges. The system has some drawbacks such as it cannot include real-time data and it is dependent on external services, but they do not diminish the effectiveness of the system in its purpose.

More enhancements will make RideWise a more complex, scalable and intelligent travel assistant, capable of significantly impacting the overall commuting experience of its users.

CHAPTER - 8

8. LIMITATIONS AND FUTURE ENHANCEMENTS

8.1. Limitations

Though the RideWise system has an effective and smart way of optimizing multi-modal travel, they have certain limitations that must be overcome.

1. The system uses estimated values of costs and travel time and therefore, the recommendations are not always realistic in regard to the real-time conditions.
2. The system relies on external services like OpenStreetMap and OSRM, and as such, any down-time or restrictions of these services may impact performance.
3. Metro and transport data: these are premised on pre-defined datasets, which fail to be always accurate and relevant to all locations.
4. Ride cost estimation excludes real-time surge pricing offered by such services as Uber and Ola, and thus the actual prices can be different.
5. The system fails to furnish user preferences like comfort, travel history or mode of transport that is preferred by the user.
6. The system primarily targets large cities, and thus the accuracy of the data in rural or less-developed regions may not be as high as possible.
7. Travel time estimation is not a complete consideration of the real time traffic congestion which could influence the efficiency of routes.
8. The system cannot support offline use since it needs an active internet connection in order to operate.
9. The performance can be lowered when serving many users without the necessary scaling in the backend.
10. The system redirects the user to other sites to make the reservations rather than finalizing the reservations in the application.

8.2. Future Enhancements

The RideWise system may be further enhanced by the introduction of more advanced features to make travel planning smarter, faster, and easier. The following are some of the improvements that could be made:

1. Incorporate real-time information of ride-hailing services, public transport, and traffic systems to enhance the quality of the route suggestions.
2. Introduce dynamic pricing used by other applications such as Uber and Ola in order to have more accurate costs estimates.
3. Have official metro and public transport APIs to give real-time schedule, routes and availability.
4. Implement user customization by taking into account such preferences like budget, comfort, travel history, and preferred modes of transport.
5. Incorporate real-time traffic analysis to recommend more effective routes and minimize travel delays.
6. Provide offline access to essential services such as saved routes and recent searches.
7. Introduce cloud-based scaling to be able to serve a great number of users with the ability to scale the system without performance problems.
8. Incorporate in-app booking capabilities to enable users to make ride or ticket bookings directly in the system.
9. Enhance the search and filter features to deliver better travel recommendations that are more specific and accurate.
10. Extend the system to cover more cities and rural locations with improved transport data coverage.

A. APPENDIX

A.1. References

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A.2. Project Timeline Table

RideWise project timeline: 03 February 2026 to 10 April 2026

Table A.1: Project Timeline – RideWise – Smart Multi-Modal Travel Optimization System

Date	What discussed	What actions taken
03-02-2026	Selection of project topic (RideWise)	Finalized project title and basic system idea
07-02-2026	Problem statement and objectives	Defined project goals and project scope
12-02-2026	Literature survey discussion	Collected and reviewed research papers
18-02-2026	Modules planning	Identified system modules and functionalities
24-02-2026	API integration methods	Planned OpenStreetMap and OSRM API usage
02-03-2026	Route optimization techniques	Studied routing and optimization methods
20-03-2026	System architecture design	Created architecture and UML diagrams
25-03-2026	Implementation planning	Started frontend and backend development
30-03-2026	API integration and testing	Integrated APIs into the system
03-04-2026	Recommendation module testing	Tested route generation and optimization process
06-04-2026	User interface design	Improved UI using HTML, CSS, and JavaScript
08-04-2026	Testing and debugging	Fixed errors and improved system performance
10-04-2026	Final documentation and report preparation	Completed project report and documentation